

Eco-friendly marine antifouling coating consisting of cellulose nanocrystals with bioinspired micromorphology

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ABSTRACT

Nanomaterial-incorporated surfaces with microstructures have been widely used for marine antifouling coatings, yet limited green antifouling coatings are currently available for sustainable application, given the potential environmental effects of nanomaterial-based nanofillers. Here, by using natural sourced nanomaterials (cellulose nanocrystals, CNCs) as nanofillers, a nanocomposite superhydrophobic coating was fabricated via a simple sol-gel synthesis method. Notably, CNCs were firstly applied in the marine antifouling realm to form uniform and stable coatings, which were condensed with hydroxyl groups of hydrolyzed tetrapropyl zirconate, 3-glycidioxypropyltrimethoxysilane, and methyltrimethoxysilane. The synthesized coatings gained a biomimetic microscopic ridge-like surface, where more CNCs contents contributed to finer microstructures. As a result of the influence of CNCs content on surface wettability and antifouling properties, the coating with CNCs accounting for 20 wt% of the total solid content (CNC20) delivered the best antifouling performance. Furthermore, 90-day marine field tests verified CNC20's excellent antifouling ability, reducing fouling by 82 % in comparison to the control group. Such a biomimicry study provides a novel strategy for the development of environmentally friendly coatings with CNCs nanofillers.

1. Introduction

Marine antifouling has gained increasing attention on account of the serious adverse effects of biofouling on maritime activities (Bloecher & Floerl, 2020; Li et al., 2020; Xie, Pan, Ma, & Zhang, 2019). While there have been impressive advances in antifouling coatings, most commercial coatings still contain toxic metals, such as cuprous oxide and zinc pyridine (Dooze, Ranke, Stock, Bottin-Weber, & Jastorff, 2004; Kristensen et al., 2010). Marine coatings consist of synthetic polymer backbones of high molecular weight, such as epoxy, rosins, acrylic polymers, among others, which are likely to be eroded and release microplastics (Turner, 2021). The biocides and microparticles are widespread across environmental compartments and accumulate in marine biota, not only threaten marine ecology, but also raise health concerns for humans (Nir & Rechtes, 2016; Torres & De-la-Torre, 2021). Therefore, nontoxic sustainable alternatives are therefore urgently needed (Han et al., 2021).

There are many creatures that gain antifouling abilities, such as

sharks, crabs, and mangrove tree *Sonneratia apetala*, which give us inspirations (Jin, Tian, Bing, Zhao, & Ren, 2022; Xia et al., 2019; Zhao et al., 2020). Currently, various approaches have been developed to fabricate micro/nanostructured surfaces, mimicking the micromorphology of natural surfaces, since their outstanding fouling prevention features and independent of biocides releasing (Chen et al., 2021; Liu & Li, 2012). The techniques include templating or soft lithography, etching processes, electrostatic approaches, mechanical micro-machining and so on (Jin et al., 2022). However, most of them are complicated, inefficient, expensive, and inadvisable for commercial applications. Incorporating nanomaterials to fabricate micro-structured surfaces has become a popular trend in recent years, since it is inexpensive and easy to modulate (Kumar et al., 2021; Pourhashem et al., 2022). What's more, nanofillers promote mechanical strength and roughness, leading to an enhancement in the durability and hydrophobicity of surfaces (Selim, El-Safty, Shenashen, Higazy, & Elmarakbi, 2020). Nonetheless, nanofillers may be released from the coatings,

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weakening the antifouling ability and negatively affecting the marine ecology. Thus, a more environmentally friendly and sustainable coating is awaiting to be developed.

Biomaterials are potential sustainable alternatives to conventional antifouling paints. Cellulose nanocrystals (CNCs) have been used as the reinforcing and multifunctional nanofillers for polymers (Ansari, Ding, Berglund, & Dauskardt, 2018; Qiao et al., 2021; Yang et al., 2022). As they were natural resourced, there is no burden on the environment with CNCs incorporating. However, to our best knowledge, there is no report on the fabrication of marine antifouling coating using CNCs. The high hydrophilicity of CNCs makes their gels or membranes strong water absorbers, whereas CNCs tend to aggregate in high-salt solutions (H. Qiu et al., 2022), which limits their long-term use in marine environment. Herein, we reason that a nanocomposite hydrophobic surface with CNCs filling will enable us not only to protect CNCs from damage but also to realize environmentally friendly antifouling. Specifically, as shown in Scheme 1, a sol-gel method was employed for facile synthesis of eco-friendly marine antifouling coating incorporating CNCs. The CNCs content of the coating was modulated to control its morphology, and subsequently a ridge-like morphology mimicking *S. apetalis* leaves was achieved. Additionally, different CNCs contents were tested for their influence on surface wettability, stability, fouling release, and self-cleaning. Moreover, the laboratory antifouling capabilities and potential for practical application in the sea environment were demonstrated.

2. Materials and methods

2.1. Materials

Sulfuric acid, tetrapropyl zirconate (TPOZ), glacial acetic acid (GAA), 3-glycidyloxypropyltrimethoxysilane (GPTMS), methyltrimethoxysilane

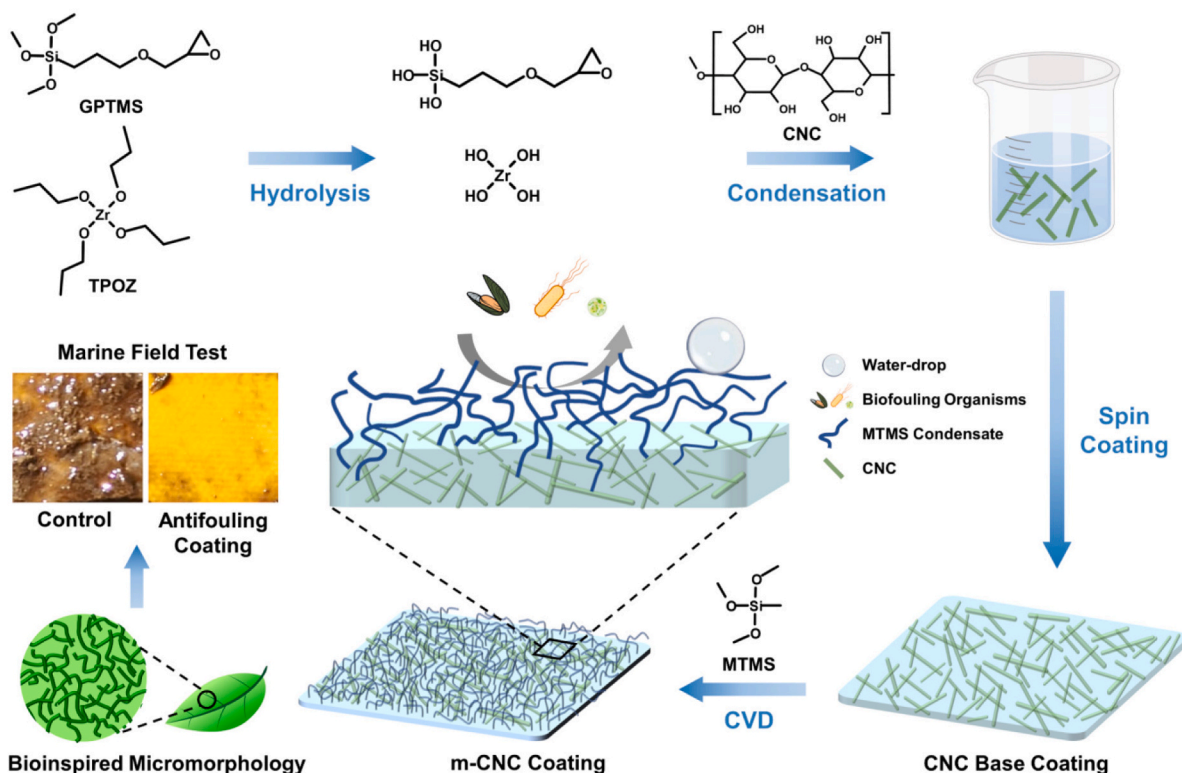
(MTMS), Isopropyl alcohol (IPA), diiodomethane, $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$, tryptone, SiO_2 micropowder, and methylene blue were purchased from Aladdin. *Escherichia coli* (*E. coli*) was purchased from China General Microbiological Culture Collection Center. *Nitzschia closterium f. minutissima* (*N. closterium f. minutissima*) and natural seawater were purchased from Shanghai Guangyu Biological Technology Company.

2.2. CNCs preparation and characterization

CNCs were prepared according to a previously reported method (Lu & Hsieh, 2010). The morphology of the CNCs was observed by atomic force microscope (AFM) (Nanosurf C3000i Controller) and transmission electron microscope (TEM) (JEM-1400Flash, JEOL). The dimension of CNCs was measured through 100 samples from TEM image by Nano Measure software. Fourier transform infrared spectroscopy (FTIR, Nicolet 6700, ThermoFisher Scientific) was used to determine the chemical structure of the CNCs. The crystallinity of CNCs was determined by X-ray diffractometer (XRD) (D8-Focus, Broker AXS, Germany) using $\text{CuK}\alpha$ radiation in the 2θ range from 5° to 45° . The crystallinity index (CI) was calculated by Segal equation (Segal, Creely, Martin, & Conrad, 1959):

$$\text{CrI} = \frac{(I_{002} - I_{\text{am}})}{I_{002}} \times 100\%$$

where I_{002} is the maximum intensity of 002 crystal plane at 22.9° and I_{am} is the intensity of amorphous diffraction, which is taken around 18° . Zeta potential and particle size of CNCs were tested by dynamic light scattering (Zetasizer Nano ZS, Malvern).



Scheme 1. Schematic synthesis route of superhydrophobic micro-structured antifouling coating. Hydrolyzed GPTMS and TPOZ first underwent condensation reactions with CNCs to form a covalent network, then a chemical vapor deposition process with MTMS was performed for silanization modification, after which the superhydrophobic coating with a microscopic ridge-like morphology mimicking *S. apetalis* leaves was produced and the coating exhibited excellent antifouling ability.

2.3. Coatings preparation

The hybrid coatings were prepared using sol-gel chemistry (Ansari et al., 2018). First, TPOZ was added to GAA and stirred for 5 min. Then, TPOZ/GAA solution was mixed with GPTMS solution (2 wt%) and stirred until a clear solution was obtained. A different amount of CNCs suspensions were added to the above solution and stirred for 2 h. CNCs account for 0, 5, 10, 20, 40, 70 wt% of the total solid content. The solution was diluted with water: IPA (1:2) to finally achieved a solid content of 0.2 wt% and applied to glass slides by spin coating. After volatilizing at ambient temperature, the coatings and polytetrafluoroethylene vials containing MTMS were placed in a sealed container, then heated at 70 °C for 3 h followed by 80 °C for 5 h to obtain silylated superhydrophobic coatings. For convenience, the as-prepared coatings were designated as CNC0, CNC5, CNC10, CNC20, CNC40, and CNC70, respectively. FTIR was used to determine the chemical constitution of the coatings. Surface elemental analysis was performed by X-ray photoelectron spectroscopy (XPS) (K-Alpha+, ThermoFisher Scientific) with Al K α radiation, and the binding energy was calibrated with C1s (284.80).

2.4. Surface characteristics measurement

2.4.1. Morphology characterization

The surface morphology of the coatings was observed by AFM and Field-emission scanning electron microscopy (FESEM) (Regulus-8100, HITACHI). The cross-section morphology and thickness of the coatings was observed by FESEM. Surface roughness tester (SJ-210, Mitutoyo) was used to further characterize surface morphology, according to ISO1997.

2.4.2. Contact angle measurement and surface free energy calculation

The CA of deionized water (WCA) and diiodomethane (DCA) on the coatings was measured by a contact angle meter (Theta Lite, Biolin Scientific). Measurements were performed on five random regions of each coating to obtain an average value. The surface free energy (SFE) was calculated by the following formula according to the Owens-Wendt-Rabel-Kaelble method (Owens & Wendt, 1969).

$$\gamma_s = \gamma_s^d + \gamma_s^p$$

$$\gamma_L(1 + \cos\theta) = 2(\gamma_s^d\gamma_L^d)^{1/2} + 2(\gamma_s^p\gamma_L^p)^{1/2}$$

For water, $\gamma_L^d = 21.8$ mN/m, $\gamma_L^p = 51.0$ mN/m. For diiodomethane, $\gamma_L^d = 48.5$ mN/m, $\gamma_L^p = 2.3$ mN/m. γ_s^d refers to the dispersion component of the SFE. γ_s^p refers to the polar component of the SFE.

2.4.3. Stability test

The superhydrophobic coatings were immersed in seawater for two months and taken out every few weeks. Before measurement, the coatings were rinsed gently with deionized water to remove the influence of the sea salt, and then blow-dried with nitrogen. The WCA was measured according to the above method.

2.5. Antifouling tests

2.5.1. Self-cleaning test

The protocol of the self-cleaning test referred to a published article (Tong et al., 2022). The coatings were contaminated by solid (CuCl₂·2H₂O, tryptone, and hydrophobic SiO₂ micropowder) and liquid (algae, methylene blue solution, and muddy dispersion). Then 100 μ L of deionized water dripped continuously to remove the solid contaminants. The inclination angle of the glass slide was about 5°.

2.5.2. Anti-biofouling test

Each coated glass slide was immersed in a sterile petri dish

containing 10 mL of *E. coli* suspension ($\sim 10^6$ CFU) and incubated at 37 °C for 12 h. After incubation, the coatings were rinsed three times with PBS buffer to remove unsteadily adhered bacteria and the rinsing fluid was discarded. Subsequently, the coatings were rinsed repeatedly to completely separate the attached bacteria. 200 μ L of PBS-resuspended bacteria from each coating was spread on Luria-Bertani agar plates and finally cultured for another 18 h at 37 °C. The antibacterial experiment was repeated three times, and the number of colonies on the plate was counted.

Guillard's F/2 culture medium was employed to cultivate *N. closterium f. minutissima* (a typical benthic marine biofouling micro-alga) in an illumination incubator at 20 °C with a 12: 12 h light-dark cycle. Each coated glass slide was immersed in 30 mL of diatom suspension ($\sim 10^8$ cells/mL). After being cultivated for a week, the coatings were picked out and the unsteadily adhered diatom was washed away by seawater. Afterward, the diatom that settled on the coatings was observed by an optical microscope (CX-23, Olympus) and the diatom density was recorded from ten random areas.

The anti-bacterial and anti-diatom rate (*R*) were calculated by the following formula

$$R = \frac{N_b - N_c}{N_b} \times 100\%$$

N_b and N_c respectively refer to the number of bacteria colonies or diatoms corresponding to the bared control group and the coated experimental group.

2.5.3. Removal test of pseudobarnacles

The test was carried out following ASTM D5618 (2011). A cylindrical aluminum ingot with a diameter and height of 1 cm was stuck on the coating by epoxy adhesive to simulate the adhesion of the barnacle. The adhesive was cured for 3 d under ambient conditions. The aluminum ingot was removed with a force gauge at a constant speed, and the pulling force was parallel to the surface. The force for the detachment of aluminum ingot from the coating was measured. For each coating, three points were tested.

2.5.4. Marine field test

The marine field tests were conducted for 90 d from September to December in the Bohai Sea (117°46'E, 39°43'N), Tianjin, China. The coated epoxy resin panels (50 mm $\times$ 50 mm $\times$ 3 mm) were immersed at a depth of 0.5 m. Three replicates for each coating were used. After a certain period, the coatings were taken out and rinsed with seawater to remove unsteadily attached fouling, then recorded by the camera. The fouling area was quantified by ImageJ software. Considering the edge effect, only the 40 mm $\times$ 40 mm area in the center of the panels was included. To further clarify the organisms attached to the panels, the fouling was removed and diluted to the same multiple in seawater. The fouling solution was observed under a microscope.

3. Results and discussion

3.1. Morphology and chemical constitution of CNCs modified coatings

The CNCs applied in this work has been characterized by TEM, AFM, dynamic light scattering, FTIR and XRD, as shown in Fig. S1. The dimension of CNCs was counted from TEM image, which showed an average length of 153.32 nm and width of 17.59 nm (Fig. S1C and D). Through dynamic light scattering method, the particle size distribution and zeta potential of -30.7 mV were attained (Fig. S1F). The crystallinity of CNCs in this work was about 76.27 % (Fig. S1H).

By employing a simple two-step approach, the coatings were first formed by a sol-gel method, and then fabricated via salinization modification to produce an excellent antifouling coating. The first step involved the sol-gel process, which was driven by two main reactions: hydrolysis and condensation, as can be seen in Scheme 1. Regarding the

FTIR spectrum (Fig. 1A), the absorption peaks at 1027 and 880 cm^{-1} were classified as Si—O bonds, containing Si—O—Si and Si—O—C bonds (Wanka et al., 2020), which arose from the condensation of hydroxyl groups generated by CNCs and GPTMS hydrolysis or GPTMS self-condensation. Besides, at 1121 cm^{-1} , the band was assigned to C—O (Ge, Huang, Zhou, & Luo, 2016), mainly made up of C—O—C and C—O—Zr bonds in condensation products (Milošev, Kapun, Rodič, & Iskra, 2015). These results confirmed that TPOZ and GPTMS were condensed with CNCs to form a covalent network (CNC base coating), which can prevent the release of CNCs and enhance the stability coatings.

Chemical vapor deposition (CVD) with MTMS for silanization modification followed by methyl modification replaced hydrophilic hydroxyl groups with hydrophobic methyl groups was applied in the second step. The spectrum of m-CNC coating appeared with two strong absorption peaks, 753 cm^{-1} and 880 cm^{-1} (Fig. 1A), which represented Si—C and Si—O bonds respectively, demonstrating the connection of MTMS and CNC base coating. The resultant CNCs modified coating was denoted as m-CNC coating. And the possible reaction between MTMS and the CNC base coating was illustrated in Fig. S2 (Guan, Cheng, & Wang, 2018).

A subsequent XPS analysis of both coatings confirmed that C, O, Si, and Zr were present (Fig. 1B). Specifically, the Zr peak was weakened after silanization modification, while the Si peak was strengthened, confirming the formation of the m-CNC coating. Fig. 1C–F showed the high-resolution spectra of elements before and after MTMS modification. The characteristic peaks and corresponding chemical bonds are marked in the figures and detailed in the supplementary information. And an estimate of chemical bond content can be derived from Table S1. In agreement with the FTIR results, the XPS also demonstrated the successful preparation of the two coatings.

After successful fabrication of designed coatings, the morphology was then investigated. As shown in Fig. 2, crosslinked polymer morphologies resembling curly ridges and grooves can be seen similar to those of *S. apetala* leaves (Fig. S3). Generally, the inter-ridge spacing of *S. apetala* leaves is approximately 5 μm , and the ridge thickness is approximately 1 μm (Fu et al., 2018). The m-CNC coating we created showed a finer microstructure, with a thickness of approximately 500 nm. A further decrease in the interridge spacing and the ridge thickness was observed with the increase of CNCs content. From the cross-section

images, we found the similar phenomenon. Besides, the coatings thickened with higher CNCs content. CNCs increased the density of hydroxyl groups on the surface of CNC base coating, so as to expose more reaction sites, then the silanization products will grow denser and slenderer on the surface. To further study the morphology of the coatings, we measured their roughness. As shown in Table 1, the surface became flat with increasing CNCs content, that was to say CNCs contributed to the homogeneous and compact ridge-like morphology. For observing CNCs distribution in the coating and its effect on the morphology, AFM was employed to characterize the CNC base coating. According to Fig. S4, all hybrid base coatings had a homogeneous interface, with needle-like CNCs uniformly dispersed. As CNCs content increased, the thickness and roughness of the coatings showed modest increase (Table 1). The relatively smooth and flat morphology indicated that the CNC base coating had nice film-forming property (Song et al., 2021). Chemical compatibility between the hydroxyl-rich CNCs and the hydrolyzed TPOZ/GPTMS prevented phase separation and allowed intimate mixing of the components (Ansari et al., 2018). The uniform distribution of CNCs and good compatibility of the coatings make them further modified to form an ideal micromorphology.

3.2. Wettability of CNCs modified coatings

In order to determine the surface wettability of the fabricated microstructure, WCA and DCA were performed. Testing of contact angles was performed manually with the help of a pipette. Fig. 3A demonstrated the strong hydrophobicity of the surface by demonstrating how no water droplet adhered to it. The WCA of all freshly prepared m-CNC coatings exceeded 150° irrespective of their CNCs content (Fig. 3B), while CNC base coatings were hydrophilic (Table 1). This proved that the surface silylation and micromorphology played a crucial role in superhydrophobicity. And a “volcanic” type tendency to CNCs content was observed, with CNC20 exhibiting the highest WCA, reaching 170° (Fig. S5). The fabricated compact and fine micromorphology led to super-hydrophobicity. The WCA firstly increased and then decreased with the increasing CNCs content. When CNCs content surpassed 20 %, the hydroxyl groups on the surface may not be completely silylated, which caused the decrease in WCA. Furthermore, DCA was also tested to explore the oil repellency and SFE of the coatings. The average DCA of the m-CNC coatings ranged between 135° and 150° (Figs. 3B and S6),

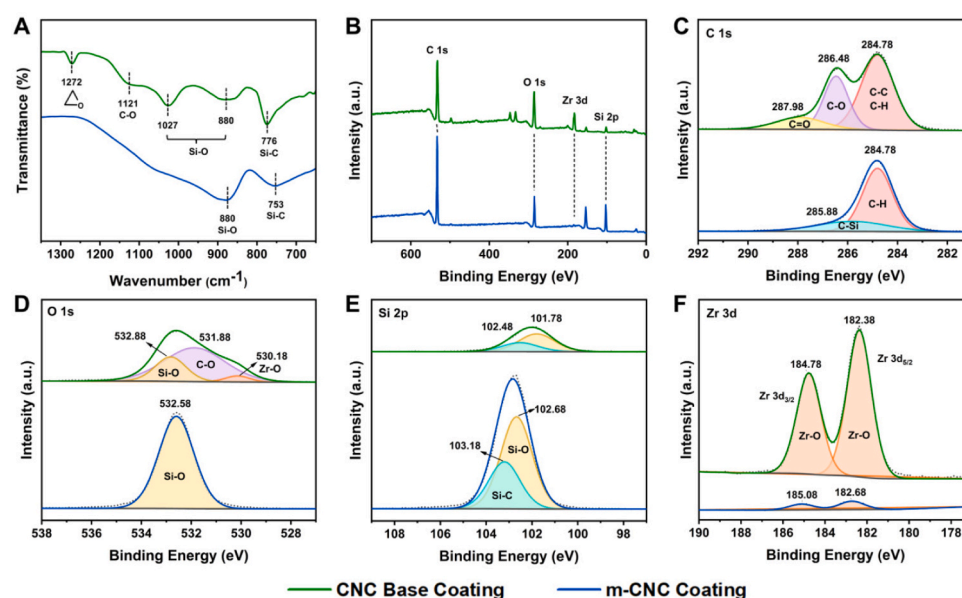


Fig. 1. Chemical constitution of coatings. (A) ATR-FTIR spectra. (B) XPS survey spectra. (C) C 1s spectra. (D) O 1s spectra. (E) Si 2p spectra. (F) Zr 3d spectra. CNC base coating is represented by the green line in the upper part of each figure, while m-CNC coating is represented by the blue line in the lower part.

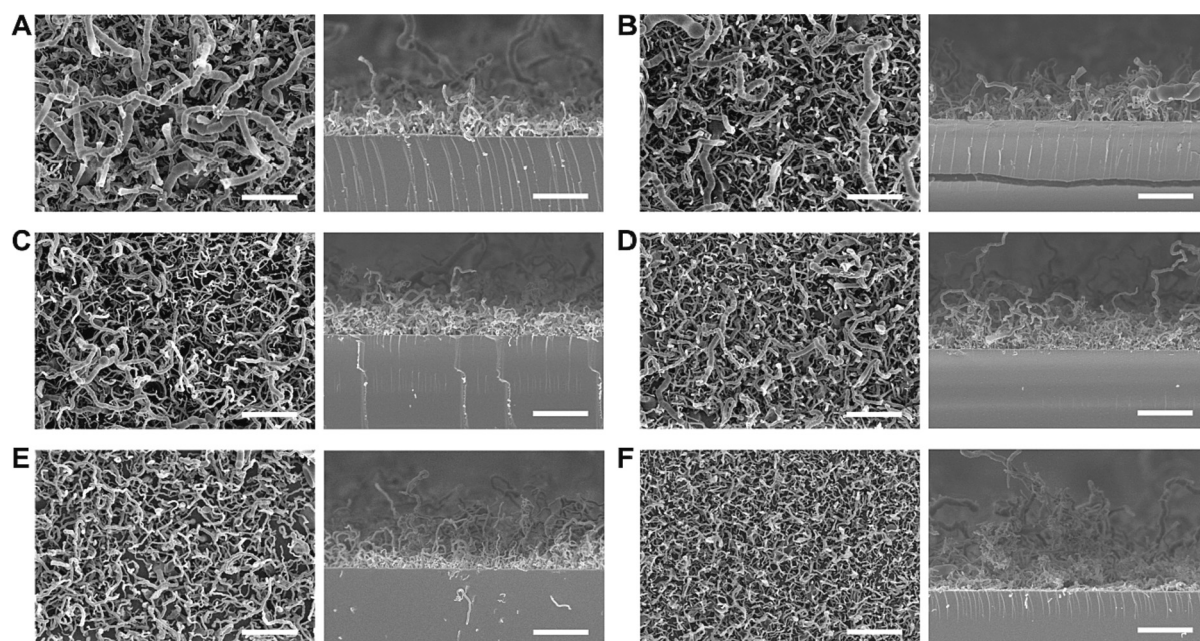


Fig. 2. Morphology characterization of coatings with varying CNCs contents (left: surface; right: cross-section). (A) CNC0, (B) CNC5, (C) CNC10, (D) CNC20, (E) CNC40, (F) CNC70. Scale bar = 5 μm .

Table 1

Surface characteristics before and after silanization modification.

CNCs Content (%)		0	5	10	20	40	70
Thickness (μm)	CNC base coating	0.47	0.49	0.53	0.58	0.62	0.71
	m-CNC coating	3.06	3.62	4.05	4.62	6.09	7.39
Roughness (R_a , μm)	CNC base coating	0.011	0.024	0.029	0.038	0.041	0.043
	m-CNC coating	0.252	0.200	0.193	0.189	0.142	0.129
WCA ($^\circ$)	CNC base coating	56.9	55.8	52.1	47.4	41.3	30.8
	m-CNC coating	155.4	153.0	158.2	162.8	158.8	153.8

indicating that these coatings were oleophobic. The super liquid repellency is mainly attributed to the micro/nanoscale hierarchical structure on the surfaces. Similarly to the WCA, the DCA increased and then decreased as CNCs content increased.

The SFE was closely related to the wettability of the surface. As shown in Fig. 3C, due to superhydrophobic and oleophobic properties, the m-CNC coatings showed extremely low SFE value (below 1 mN/m). In the previous works, the commercial low SFE coatings, such as polydimethylsiloxane (PDMS) and polyurethane, have the SFE of about 20 mN/m (Qiu et al., 2018). Some nanostructured surfaces have the SFE of about 10 mN/m (Boo, Lee, & Elimelech, 2016; Selim, Fatthallah, Higazy, Hao, & Mo, 2022). Low SFE of our coating represents small interaction between coatings and contaminants, which prevents fouling adhesion (Moghadam, Parsimehr, & Ehsani, 2021).

In order to ensure a long-term antifouling performance of the coatings in the harsh marine environment, long-term stability is essential. Observing the coatings' hydrophobicity over time was carried out by immersing them in seawater (Fig. 3D and Table S3). After two weeks of immersion, the WCA of CNC0, CNC40, and CNC70 decreased sharply while CNC10 and CNC20 sustained their superhydrophobicity. After immersion for another two weeks, the WCA of CNC0 was still significantly lowered, compared with coatings incorporated with CNCs. This phenomenon suggested that incorporating the appropriate concentration of CNCs can promote the stability of the coatings. Excessive CNCs will not only reduce the hydrophobicity of the coating, but also easily absorb water and swell, affecting the stability. During the second month, the coatings' WCA decreased by five degrees, indicating that the wettability of the surface gradually stabilized. In all cases, the WCA of

the coatings was above 130° , and the coatings still exhibited good hydrophobicity. Incredibly, CNC20 had an average WCA of almost 150° , ensuring that it maintained its superhydrophobic properties. The statistical analysis showed that the difference between CNC20 and CNC0 was significant, indicating that the proper CNCs content had a positive effect on the stability of the coatings. Overall, a certain degree of hydrophobicity was maintained by the coatings in seawater, which contributed to their long-term antifouling capacity.

Hydrophobicity mainly depends on surface chemistry and morphology (Moghadam et al., 2021). The rich silane groups and ridge-like microstructure on the surface of m-CNC coatings endow them superhydrophobicity and good stability. To further determine the mechanism of superhydrophobic surface formation, the coating was immersed in deionized water and the interface between liquid and solid was observed using an optical microscope. As presented in Fig. S7, bubbles were found at the interface. In particular, bubble numbers increased with CNCs content, while volume decreased, which was associated with coating surface morphology. The smaller size of micromorphology created by CNCs content allows for improved air trapping in grooves and ridges since the solid-liquid interface is enhanced. As air is a superhydrophobic feature with a WCA of 180° , this phenomenon well explained the excellent superhydrophobicity (Selim et al., 2020). Additionally, entrapped air can reduce the actual contact area between a superhydrophobic surface and organisms, inhibiting biofouling attachment (Arnott, Wu, Vucko, & Lamb, 2014; Truong et al., 2012). After 15 min, some bubbles became smaller or broke. More bubbles were retained by CNC20 than CNC10 and CNC40, suggesting that CNC20 has excellent air capture and retention capabilities, which

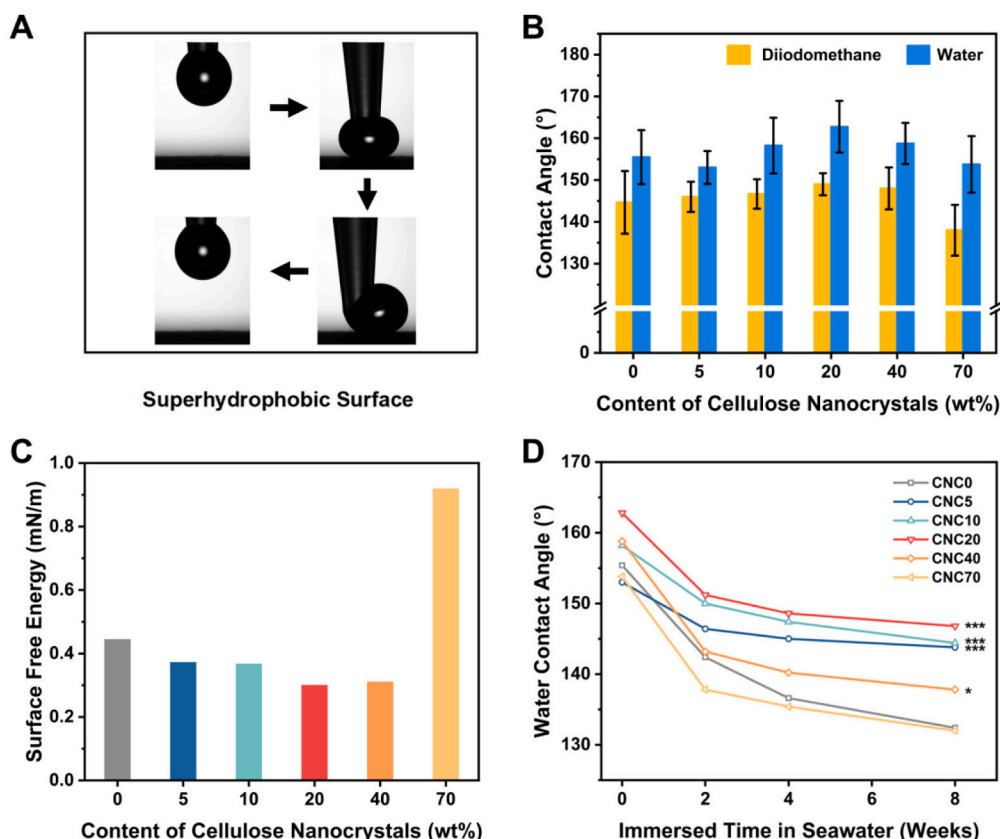


Fig. 3. (A) A drop of water on the CNC20 surface. (B) WCA and DCA of fresh-prepared coatings. (C) SFE calculated by average WCA and DCA. (D) Time-dependent variation in WCA in seawater. One-way ANOVA was carried out to assess the differences of each coating with CNC0 after eight weeks immersion. Statistical significance was set as follows: * $P < 0.01$, *** $P < 0.001$.

can explain its best hydrophobicity and stability and potentially making it more foul-resistant.

3.3. Marine antifouling performance

3.3.1. Self-cleaning performance towards abiotic contaminants

The marine environment can pollute the coating with solids and micromolecules, such as silt and dye. The coating should therefore be tested for its resistance to fouling by examining its self-cleaning property. In Figs. 4 and S8, several solids and liquids that may occur in the sea were chosen, and the self-cleaning properties of coatings with different CNCs content did not differ significantly. As for solid contaminants, $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ and tryptone dissolved in water and slid out of the coating with water droplets, while hydrophobic SiO_2 micropowder was directly washed away by water droplets. Because of the coating's micromorphology and low surface energy, solid contaminants had a small contact area and weak interaction with the coating, so water droplets readily carried the solids down. Liquid contaminants such as algae, methylene blue solution, and muddy dispersion were able to slide down instantly without residue, as seen in Movies S1–S6. It was believed that the superhydrophobicity was responsible for the excellent liquid repellency. In the motion of rolling over the surface, the drops of water formed spherical droplets and carried contaminants away with them. Such self-cleaning phenomenon, which is known as the lotus effect (Barthlott & Neinhuis, 1997), is caused by a combination of low surface energy and microstructure on the surface. As a result, the fantastic self-cleaning performance of the coating contributes to the resistance of fouling in the sea.

3.3.2. Anti-biofouling assessment

The presence of bacteria and diatoms on the surface is critical to biofouling. These organisms can form a biofilm, which facilitates macrofouling organism colonization. To determine the coatings' antifouling properties, a laboratory test for adhesion to bacteria and diatoms was conducted before evaluating their performance in the marine field. Due to *E. coli*'s widespread presence in seawater, it was chosen to assess the m-CNC coating's antibacterial ability by plate counting photographs of *E. coli* colonies recultivated after being rinsed from coatings were shown in Fig. 5A. Clearly, every coating was effective in preventing bacterial growth, decreasing by >60 % the number of colonies. Particularly, the antifouling performance of the coatings was roughly proportional to their WCA, where antifouling ability increased with increasing WCA. As a result, CNC20, with the highest WCA, showed the best antibacterial action, up to 95 % (Fig. 5C and E).

For assessing the anti-diatom capabilities of the m-CNC coating, *N. closterium f. minutissima*, a typical marine biofouling microalga, was chosen. After one week of immersion in diatom suspension of an exponential growth phase, coatings were tested for anti-diatom performance. A comparison of surfaces of coatings after immersion in diatom suspension for 1 d and 7 d is shown in Fig. 5B. The bare glass slide had a large number of diatoms attached after only one day of immersion, while coated glass slides did not have as many diatoms attached. As the slide remained immersed for 7 d, more diatoms grew on it. It was especially evident for the bare glass that the diatoms aggregated or even changed shape and died. As the slide remained immersed for 7 d, more diatoms grew on it. It was especially evident for the bare glass that the diatoms aggregated or even changed shape and died. In comparison to bare glass, the coated glass accumulated much fewer diatoms, with a >90 % reduction in diatom density (Fig. 5D). Interestingly, the anti-diatom

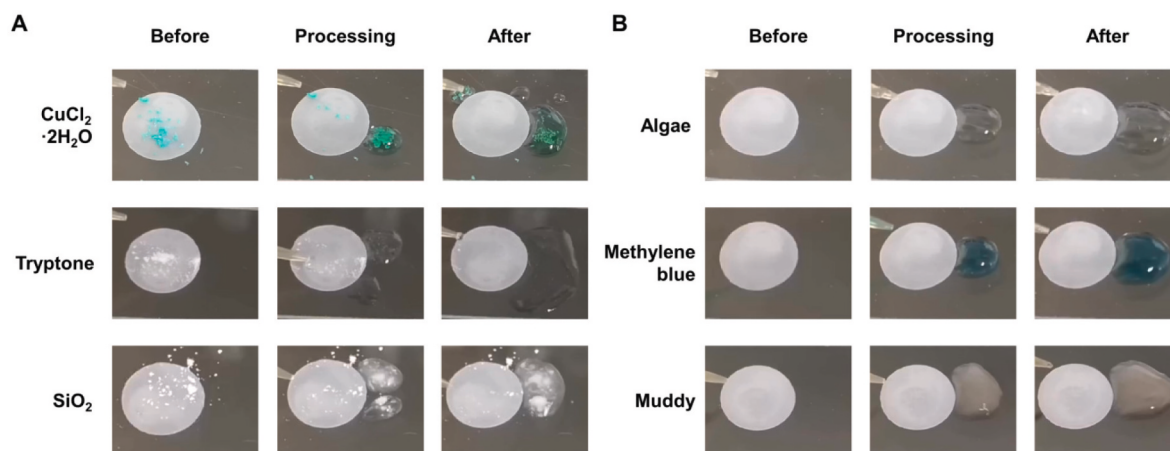


Fig. 4. Self-cleaning performance of CNC20. (A) Solid model contaminants. (B) Liquid model contaminants.

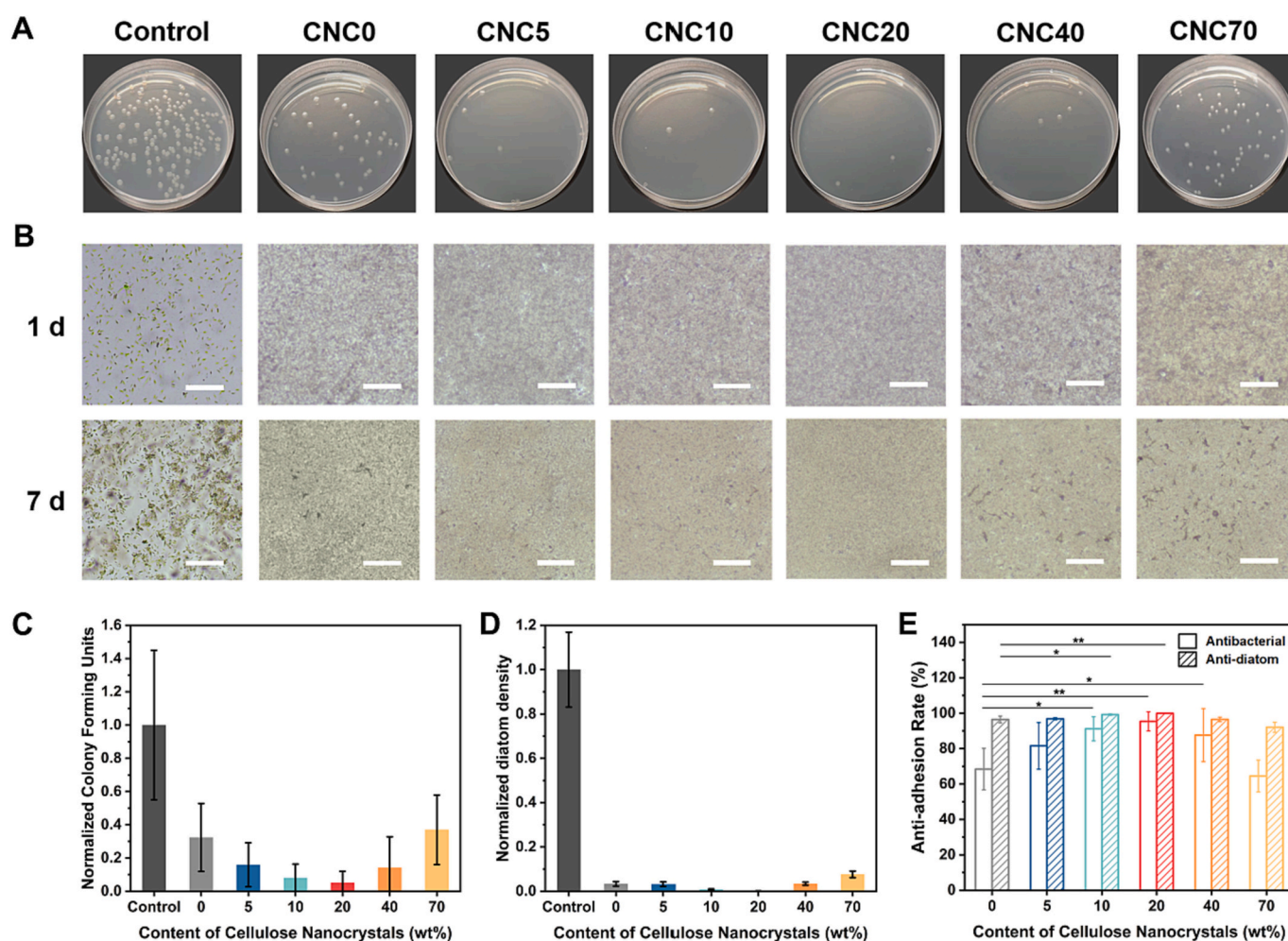


Fig. 5. Anti-biofouling assessment. Antibacterial test: (A) Images of agar plates (C) statistics of colonies on agar plates. Anti-diatom test: (B) Microscopy images of diatom adhered on the surface of coatings. Scale bar = 100 μ m. (D) Statistical chart of diatom density after 7 days' immersion. (E) Antibacterial rate and anti-diatom rate calculated by (C) and (D). One-way ANOVA was carried out to assess the two-group differences. Statistical significance was set as follows: * $P < 0.05$, ** $P < 0.01$.

performance of these coatings increased with increasing WCA, as did the antibacterial tendency. As a result, CNC20, which had the highest WCA, also showed the best anti-diatom performance, up to 98 % (Fig. 5E). Generally speaking, microstructured superhydrophobic coatings perform well against diatoms, according to literature surveys. In spite of

this, CNC20 still exhibited outstanding resistance to diatoms (Table S2). As shown in Fig. 5E, compared with CNC0, the antibacterial and anti-diatom rate of CNC20 were obviously improved, suggesting that CNCs were essential in the antifouling coatings. CNCs can regulate the surface micromorphology and enhance the stability of the coatings, yielding

better antifouling performance. With regard to Fig. 2, the more uniform and denser morphology of CNC20, without aggregation and cracking, may greatly contribute to its best antifouling performance. Together, the m-CNC coatings displayed excellent antibacterial and anti-diatom properties.

3.3.3. Fouling release behavior

Under shear force, fouling organisms can be detached from fouling release coatings, contributing to marine antifouling. By removing pseudobarnacles, fouling release ability was evaluated (Fig. S9), and lower removal strengths indicated better performance (R. Chen, Xie, Zeng, Ma, & Zhang, 2020). Comparing the removal strength of our coatings with PDMS, a commercial coating with excellent fouling release performance, it was found that CNC0 had the lowest removal strength, behaving better than PDMS. With the increase of CNCs content in the coating, the removal strength had a modest rise, but their fouling release ability was still comparable to that of PDMS. As a result, the coatings are not only capable of preventing fouling from attaching but also of separating fouling that has adhered to them.

3.3.4. Marine field antifouling assessment

The indoor antibacterial and anti-diatom tests are convenient methods to preliminarily evaluate the antifouling performance of coatings. However, the actual marine environment is extremely complex, including thousands of fouling organisms and inorganic substances. A marine field test is therefore the most direct and effective way to assess antifouling performance in practice. It is depicted in Fig. S10 that the marine field test was conducted in a severely fouled environment. CNC20, which performed best in laboratory antifouling tests, was selected for marine field testing. For comparison, CNC0, CNC10, CNC40, as well as the commercial coating (JOTUN SeaForce 60), were also applied. Each time samples were taken out, and the fouling was gently rinsed by seawater to remove the unstable adhesion. The fouling coverage area of the coating was assessed in order to represent the coating's resistance to various fouling. The surfaces of the panels after immersion in the Bohai Sea are shown in Fig. 6, while Image J software has been used to process the photos in Fig. S11. After 21 days of immersion, bare panels are slightly fouled, while coated panels are clean except for a small amount of fouling around the edges. On day 45, the fouling coverage area of the control group was nearly 50 %, which was much larger than that of the coated group. Coatings containing different CNCs content showed similar antifouling effects, reducing fouling coverage by 65 %. After 90 days of immersion, the bare panels developed a thick layer of fouling, mostly silt and seaweed, covering around 65 % of the surface. Comparatively, fouling on coatings

decreased by over 50 %, indicating that in real marine environments they were effective against fouling. Among them, CNC20 performed the best with an 82 % reduction in fouling area, which was in agreement with the results of the indoor antifouling tests. Furthermore, CNC10 and CNC20 performed even better than the commercial coating in terms of antifouling.

To further clarify the organisms attached to the panels, the fouling removed from the panels was observed under a microscope and the images were shown in Fig. S12A. It was found that a great deal of algae of different species and forms were present in the control group. As for the commercial coating, there were some green acellular microorganisms, which might be dead and broken algae, since the cuprous ion contained in the coating could kill organisms attached to its surface. Superhydrophobic m-CNC coatings, on the other hand, only contained a few microorganisms similar to *Chlorella*. Based on the results of the analysis of the visual field, the amount of microbial biomass was calculated. As shown in Fig. S12B, superhydrophobic m-CNC coatings displayed a fouling density of 87 % and 20 % lower than their control and commercial counterparts. Together, these results revealed that superhydrophobic coatings were effective in preventing adhesions and fouling.

4. Conclusions

In summary, by employing the green nanomaterials CNCs as nano-fillers, we have successfully fabricated green marine antifouling coatings with micro/nanoscale hierarchical structures via a simple sol-gel method. The mangrove tree leaves like morphology, curly ridges, and grooves, on the surface of the coating caused superhydrophobic properties. By dominating the CNCs content, the coatings displayed a more homogeneous micromorphology with smaller ridges and grooves, which improved hydrophobicity. With their hydrophobic microstructure, low surface energy, and high stability, CNCs modified coatings exhibited excellent anti-adhesion, self-cleaning, and fouling release properties, especially for CNC20, making them suitable for marine environments. Additionally, the laboratory antifouling against bacteria and diatoms showed an effective prevention of early-stage biofouling. Studies of antifouling capability were further extended to the actual sea area, proving its effectiveness against silt and marine microorganisms in 90 days. The antifouling property of CNC20 was even superior to that of commercial one. Other kinds of natural nanoscale polymers such as cellulose nanofibers and nano chitin can also be applied and tested in marine antifouling, which may be used for establishing nano/micro-structured antifouling surfaces. In addition, half of negative charges of CNCs can be modified to positive charges, which transforms CNCs into

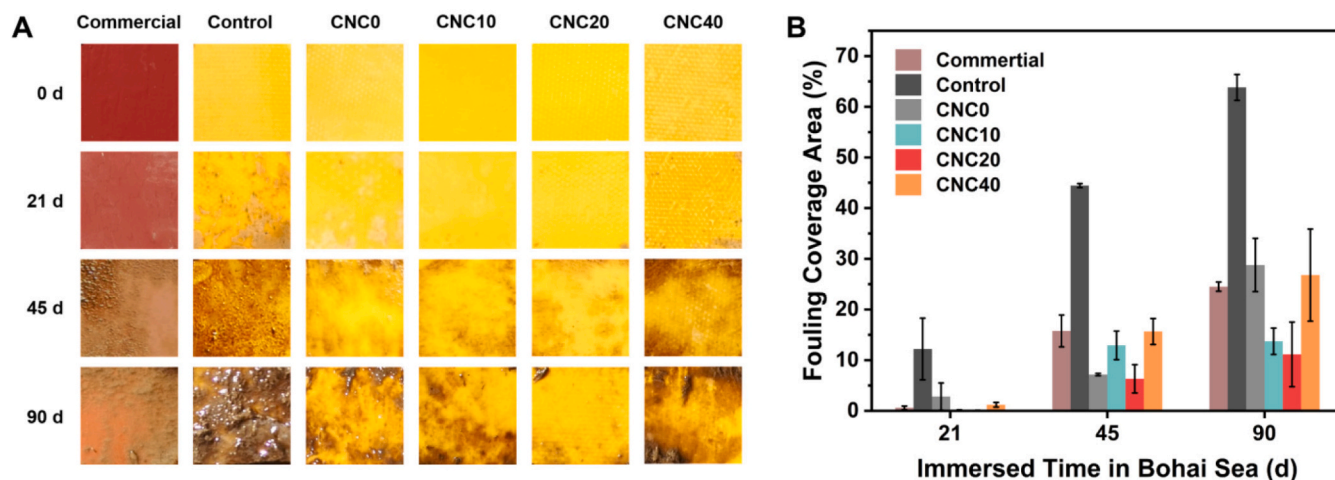


Fig. 6. Marine field assessment. (A) Images of panels, and (B) Statistics of fouling coverage area after immersed in Bohai Sea for 21, 45 and 90 days.

zwitterionic materials for developing hydrophilic antifouling coating. Not only is this work a proof of principle that CNCs can be used to fabricate marine antifouling coatings, but it also provides an eco-friendly and economical solution for marine antifouling.

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CRediT authorship contribution statement

Yanyi Duan: Investigation, Visualization, Writing – original draft. **Jiangjiexing Wu:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Wei Qi:** Writing – review & editing. **Rongxin Su:** Conceptualization, Methodology, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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